Block 4 – Network Virtualization & Cloud Computing

S2: Cloud Computing

Arquitectura de Xarxes

Universitat Pompeu Fabra

1. From Virtualization to Cloud
2. Fundamentals of Cloud Computing
3. Cloud & Container Platforms
4. Cloud Network Architecture
5. Cloud Network Services
6. Session Summary

Outline

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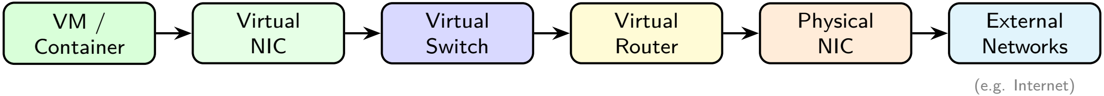
If We Can Virtualize Everything... Why Not Sell It?

Virtualization is the technology.
Cloud computing is the business model.

Learning objectives:

1. Understand cloud computing's definition and essential characteristics
2. Distinguish IaaS (Infrastructure as a Service), PaaS (Platform as a Service), and SaaS (Software as a Service) and identify key cloud platforms
3. Design a cloud virtual network with subnets, route tables, and security controls
4. Identify essential cloud network services: load balancers, proxies, and VPNs

Recap – What We Virtualized (Session 1)



- **Compute**: VMs share physical hardware via hypervisors; containers share the host kernel directly
- **Networking**: Virtual NICs, Switches, and Routers replicate the physical stack in software
- **Isolation**: VLANs and overlay networks separate tenants on shared infrastructure

The Leap to Cloud

- Virtualization = technology (**how** you run multiple workloads)
- Cloud = business model (**what** you sell to customers)

Virtualization	**Cloud Computing**
Run multiple VMs/containers on one host	Offer compute as a **service**
Internal IT optimization	External/internal consumption
Manual provisioning possible	**Automated, self-service**
You own the hardware	Provider owns the hardware

Key difference: **automation and self-service** (no phone calls, no tickets, no waiting).

Discussion: From Virtualization to Cloud

Your company already runs VMs on its own servers. Why would you move to the cloud instead of keeping everything in-house?

Hints: think about CapEx vs OpEx, scaling speed, and who manages what.

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Cloud Computing – Definition (NIST)

*A model for enabling ubiquitous, convenient, **on-demand network access** to a shared pool of configurable computing resources that can be rapidly provisioned and released with minimal management effort.*

- **Ubiquitous**: anywhere, any device
- **On-demand**: no tickets, provision instantly
- **Shared pool**: many tenants, same hardware
- **Configurable**: choose CPU, RAM, storage, network
- **Rapidly provisioned**: up and running in seconds
- **Minimal management**: provider handles the hardware

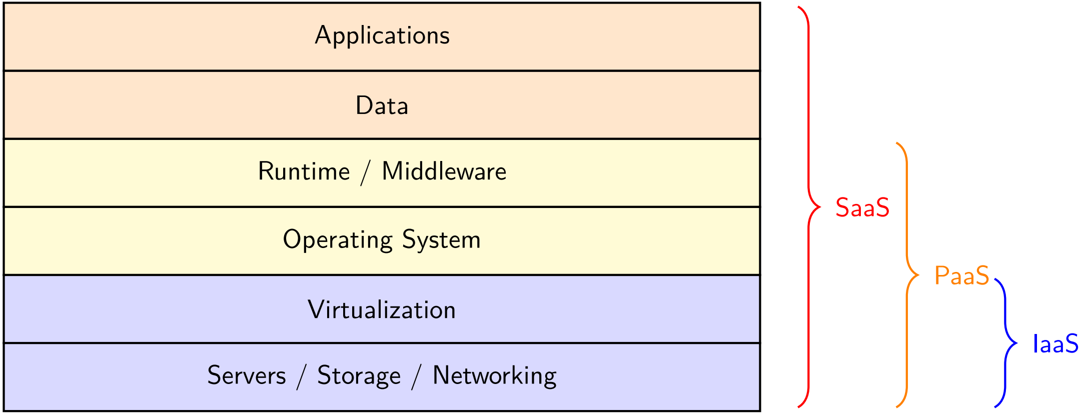
Source: NIST SP 800-145 (2011). Still the standard definition used industry-wide.

Five Essential Characteristics

Characteristic	What it means
On-demand self-service	Provision resources via portal/API, no human interaction
Broad network access	Available from any device over the network
Resource pooling	Shared infrastructure, multi-tenant
Rapid elasticity	Scale up/down automatically (e.g. Black Friday $\rightarrow$ 10$\times$)
Measured service	Pay only for what you use (CPU-hours, GB stored)

- **All five** must be present to qualify as "cloud computing"

Cloud Service Models – The Stack



IaaS, PaaS, SaaS – In Practice

- **IaaS** (AWS EC2, Azure Virtual Machines, Google Compute Engine):
  - Provider gives you VMs, storage, and networks; you manage everything above
  - “Give me a Linux server in Frankfurt, 4 CPUs, 16 GB RAM, right now”
- **PaaS** (Google App Engine, AWS Elastic Beanstalk, Azure App Service):
  - Provider gives you the runtime and deployment pipeline; you push code and data
  - “I wrote a Python web app; just run it, I do not care about servers”
- **SaaS** (Google Workspace, Microsoft 365, Salesforce, Zoom):
  - Provider gives you a complete, ready-to-use application; you manage your data only
  - “I need email for 500 employees; I do not want to run a mail server”

Service Models – Who Manages What?

Component	IaaS	PaaS	SaaS
Applications	You	You	Provider
Data	You	You	You
Runtime / Middleware	You	Provider	Provider
Operating System	You	Provider	Provider
Virtualization	Provider	Provider	Provider
Hardware	Provider	Provider	Provider
Network Services (LB, Proxy, VPN)	Provider	Provider	Provider

- IaaS → SaaS: **less control, less responsibility**
- Major providers (AWS, Azure, GCP) offer all three models

Discussion: Cloud Fundamentals

A university wants to offer a web-based tool for students to submit assignments. Should they choose IaaS, PaaS, or SaaS? Why?

Hints: consider the IT team size, maintenance burden, and how much control they need.

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Putting It All Together

What we have seen (Session 1)

- Virtual Switches and Virtual Routers
- Virtual NICs and VLANs
- Hypervisors (KVM, VMware) and containers

These are the **building blocks** of any virtual network; they do not run in isolation.

Platforms manage these building blocks

- Provision and connect VMs or containers automatically
- Handle networking, storage, and scaling under the hood
- Hide hardware complexity from the developer

Two families:

- **Cloud platforms** (AWS, Azure, GCP, OpenStack): full infrastructure stack
- **Container orchestration** (Kubernetes): deploy and scale containerized workloads

Cloud Platforms

- **Self-managed / Private Cloud**: you own or rent the servers, install and manage the virtualization software yourself. Full control, but requires dedicated staff and expertise.

Platform	Description	Logo
OpenStack	Open-source, widely used in private clouds	openstack
VMware vSphere	Enterprise leader, proprietary	vmware
Proxmox VE	Open-source, lightweight alternative	PROXMOX

- **Managed / Public Cloud**: rent capacity on demand, pay-as-you-go. Provider handles hardware, networking, cooling, security, and updates.

Provider	Description	Logo
Amazon Web Services	Market leader since 2006	aws
Microsoft Azure	Strong enterprise integration	Azure
Google Cloud Platform	Data and AI focus	Google Cloud

These map directly to the IaaS model we just covered: the provider (or you) manages compute, storage, and networking.

Public Cloud vs Private Cloud: What You Manage

Component	Public Cloud	Private Cloud
Applications	You	You
Data	You	You
Runtime / Middleware	You (IaaS) / Provider (PaaS)	You
Operating System	You (IaaS) / Provider (PaaS)	You
Virtualization	Provider	You
Hardware	Provider	You
Networking	Provider	You
Cooling / Power	Provider	You

Public cloud trades control for simplicity; private cloud gives full control at the cost of operational overhead.

Container Orchestration Platforms

- **Self-hosted / Private**: you install and operate on your own servers

Platform	Description	Logo
Docker	Container runtime; works well on a single host Managing hundreds of containers across hosts? Unmanageable	docker
Kubernetes (K8s)	Created to solve this: scheduling, scaling, networking, self-healing	kubernetes
OpenShift	Enterprise K8s by Red Hat; adds security, CI/CD, UI	Red Hat OpenShift

- **Hosted / Public**: cloud providers offer the same platforms as managed services. Each provider uses different names for essentially the same product (e.g., AWS ECS/EKS, Azure AKS, Google GKE, OpenShift on all major clouds).

Discussion: Platforms

A startup with 5 engineers needs to deploy
a web application that may go from 100 to 100,000 users.
Would you build a private cloud or use a public one? Why?

Hints: team size, upfront cost, time to market, scaling needs, control over infrastructure.

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Virtual Networks in the Cloud

- Every cloud provider lets you create your own **isolated virtual network**
  - AWS: Virtual Private Cloud (VPC)
  - Azure: Virtual Network (VNet)
  - GCP: VPC Network

- Think of it as your **private data center network** in the cloud

- You define:
  - **Address space** (e.g., `10.0.0.0/16`)
  - **Subnets** (subdivisions of the address space)
  - **Routing rules** and **security policies**

Key: under the hood, cloud providers use **overlay networks** (covered in Block 5) to isolate tenants at scale.

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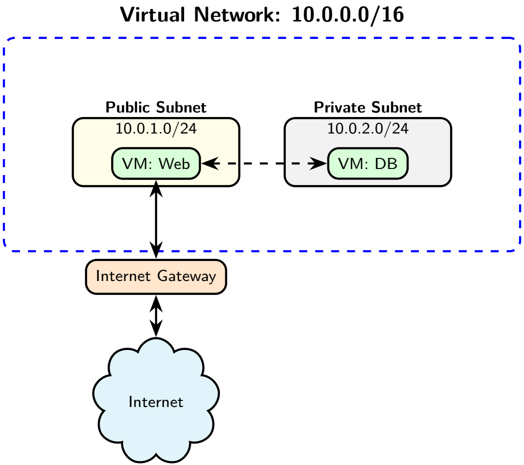
[1] AWS, "Amazon VPC User Guide"; Azure, "Virtual Network Documentation"; GCP, "VPC Documentation."

Virtual Network Architecture



Subnets: Public vs Private

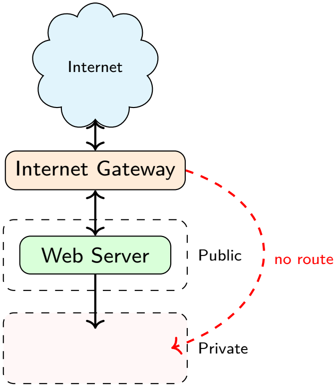
Cloud providers distinguish between subnets reachable from the Internet and isolated ones. Web servers need public access; databases and backends must stay hidden.

Public subnet:

- Route to an **Internet Gateway**
- Instances can have **public IPs**
- Use: web servers, load balancers

Private subnet:

- **No route** to the Internet
- Only **private IPs**
- Use: databases, backends

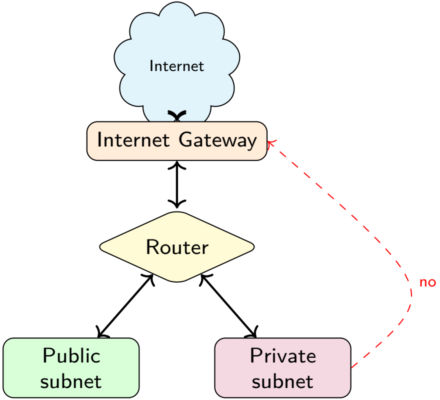


Route Tables

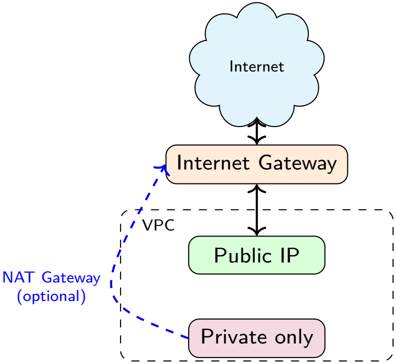
Every virtual network has an implicit **virtual router** managed by the provider. Route tables tell it where to forward traffic.

Destination	Next Hop	Subnet
10.0.0.0/16	local	Public + Private
0.0.0.0/0	Internet Gateway	Public only

- **Local route**: traffic stays inside the virtual network
- **Default route** (0.0.0.0/0): public subnets exit via the Internet Gateway
- Private subnets have no default route



Internet Gateway



In a cloud virtual network, instances are **isolated by default**. The Internet Gateway is the controlled entry and exit point.

- **Outbound only**: instance can reach the Internet, but is not reachable from outside
  - No public IP needed; the provider performs NAT transparently
- **Bidirectional**: instance is also reachable from the Internet
  - Requires a **public IP**
  - Use: web servers, load balancers

Note: a NAT Gateway can be added for private instances that need outbound Internet access (e.g. updates), without exposing them inbound.

Security Groups: Instance-Level Firewall

Once traffic reaches an instance, you still need to control what it can send and receive.

- **Security group** = virtual firewall on an instance
- Specifies allowed **inbound/outbound** traffic
- **Stateful**: allow inbound → response automatically allowed

Dir.	Proto	Port	Src/Dst
In	TCP	80	0.0.0.0/0
In	TCP	443	0.0.0.0/0
In	TCP	22	10.0.0.0/16
Out	All	All	0.0.0.0/0

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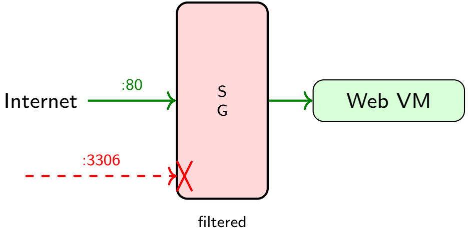


By default: all inbound denied, all outbound allowed.

[2] AWS, "Security Groups for Your VPC," docs.aws.amazon.com.

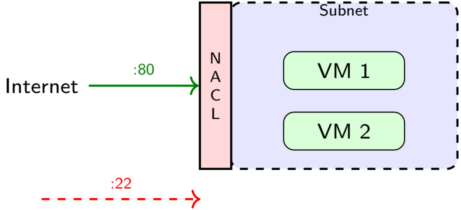
Network ACLs (Access Control Lists): Subnet-Level Firewall

Network ACLs add a layer of control at the **subnet boundary**, before traffic reaches any instance.

- **Network ACL** = firewall at the **subnet** level
- Rules evaluated in **order** (numbered, first match wins)
- **Stateless**: must explicitly allow inbound AND outbound

#	Dir.	Proto	Port	Action
100	In	TCP	80	ALLOW
200	In	TCP	443	ALLOW
*	In	All	All	DENY

[3]



Filter applies to **all** instances in the subnet.

[3] AWS, "Network ACLs," docs.aws.amazon.com.

Security Groups vs Network ACLs

	Security Groups	Network ACLs
Scope	Instance level	Subnet level
State	Stateful	Stateless
Rules	Allow only	Allow + Deny
Evaluation	All rules evaluated	First match wins
Default	Deny all inbound	Allow all

Defense in depth: use both together.

- Security Groups → fine-grained per-instance control
- Network ACLs → broad subnet-level guardrails

Discussion: Cloud Networking

You deploy a web application in your cloud virtual network. The web server needs to be public, but the database must not be reachable from the Internet. How?

Hints: think about public vs private subnets, security groups, and NAT gateways.

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Cloud Network Services: Overview

Beyond virtual networks and security controls, public cloud providers offer managed network services that simplify common infrastructure tasks, such as:

1. **Load Balancer (LB)**
2. **Proxy**
3. **Reverse Proxy**
4. **VPN (Virtual Private Network)**

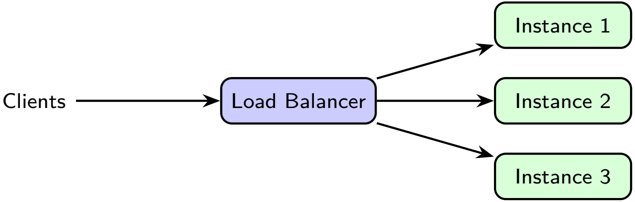
These services run as **managed offerings**: no servers to install or maintain.

Note: Proxy, Reverse Proxy, and VPN will be covered in depth in Block 6. Here we introduce them briefly.

Private cloud: these services are **not** provided automatically; you must deploy and manage them yourself (e.g. HAProxy for load balancing, OpenVPN for VPN).

Load Balancer

- Distributes incoming traffic across **multiple instances** (VMs or containers)
- Prevents any single instance from being overwhelmed
- If one instance fails, traffic is **redirected** to healthy ones



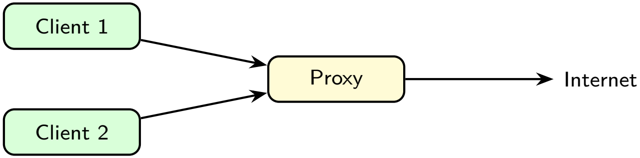
- **L4 (Transport Layer)** Load Balancer: routes based on IP and port (fast, simple)
- **L7 (Application Layer)** Load Balancer: routes based on HTTP headers, URL path, cookies (smarter)
- Cloud examples: AWS ALB/NLB, Azure Load Balancer, GCP Cloud Load Balancing

Source: AWS, "Elastic Load Balancing Documentation," docs.aws.amazon.com.

Proxy

Without a proxy, every instance in your network reaches the Internet directly: no visibility, no control. A proxy sits in front of clients and mediates all outbound traffic.

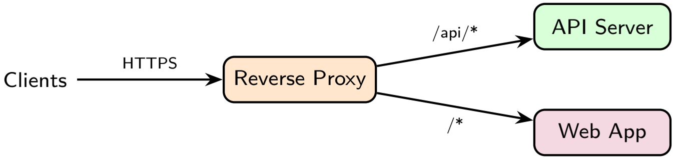
- Clients send requests to the proxy, which forwards them to the Internet



- **Access control**: block certain websites or domains
  - Security: block known malicious domains
  - Company policy: restrict access to non-work-related content
- **Caching**: store frequently accessed content, reduce bandwidth
- **Logging**: monitor what employees or VMs access
- Examples: Squid, corporate firewalls with proxy mode

Reverse Proxy

A reverse proxy sits in front of backend servers and mediates all inbound traffic. Clients never communicate directly with the backend.

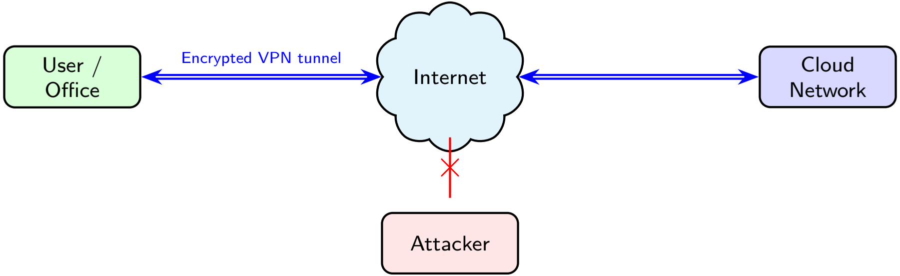


- **TLS termination**: handles HTTPS, backends receive plain HTTP
- **URL routing**: directs requests to the right backend service
- **Hides backend servers**: clients only see the proxy's address
- Examples: NGINX, HAProxy, AWS CloudFront, Azure Front Door

Will be covered in depth in Block 6.

VPN – Virtual Private Network

- A **VPN (Virtual Private Network)** creates a **secure, encrypted tunnel** over the public Internet
- Connects remote networks or users as if they were on the same private network



VPNs will be covered in depth in Block 6.

Discussion: Cloud Network Services

You are deploying a web application.
When does it make sense to add a Load Balancer?
When would you not need one?

Hints: think about the number of users, availability requirements, and what happens if your single instance goes down.

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Key Takeaways

1. Virtualization is the **technology**; cloud is the **business model**
2. Five NIST characteristics define true cloud computing
3. **IaaS / PaaS / SaaS** differ in who manages what
4. Cloud and container **platforms** deliver these models at scale
5. A **virtual network** is your isolated cloud network (CIDR, subnets, route tables)
6. **Security groups** (instance) + **ACLs** (subnet) = defense in depth
7. **Load balancers**, **proxies**, **reverse proxies**, and **VPNs** are essential cloud network services

References

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